GOVERNMENT ENTERPRISE SPECIFICATION

06 Data Schemas & Ledger Models

PostgreSQL DDL, Go Chaincode & Pydantic Contracts • GovPulse Technical Architecture Suite • Version 1.0 (Production)

Data Schemas, Ledger Data Models & API Contracts

Project: GovPulse Autonomous Digital Government Platform

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1. PostgreSQL Relational Data Schemas (DDL with Row-Level Security)

```
-- Enable UUID and Cryptographic Extensions
CREATE EXTENSION IF NOT EXISTS "uuid-ossp";
CREATE EXTENSION IF NOT EXISTS "pgcrypto";

-- 1. Citizens Identity Table (Encrypted at Rest)
CREATE TABLE citizens (
    citizen_id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    national_identity_hash VARCHAR(64) UNIQUE NOT NULL, -- SHA-256(Aadhaar/NationalID
    full_name_encrypted BYTEA NOT NULL,                 -- AES-256-GCM Encrypted
    dob_encrypted BYTEA NOT NULL,
    address_encrypted BYTEA NOT NULL,
    phone_hash VARCHAR(64) NOT NULL,
    biometric_embedding_id VARCHAR(64),                 -- Qdrant Vector Point ID
    created_at TIMESTAMP WITH TIME ZONE DEFAULT NOW(),
    updated_at TIMESTAMP WITH TIME ZONE DEFAULT NOW()
);

-- 2. Officers Registry Table
CREATE TABLE officers (
    officer_id VARCHAR(32) PRIMARY KEY,
```

```
    full_name VARCHAR(128) NOT NULL,
    department VARCHAR(64) NOT NULL,
    jurisdiction_district VARCHAR(64) NOT NULL,
    cadre_rank VARCHAR(32) NOT NULL,                    -- e.g., "GAZETTED_OFFICER_GR
    pki_public_key TEXT NOT NULL,                       -- ECDSA Public Key (X.509 PE
    active_caseload INT DEFAULT 0,
    max_caseload INT DEFAULT 50,
    is_active BOOLEAN DEFAULT TRUE,
    created_at TIMESTAMP WITH TIME ZONE DEFAULT NOW()
);

-- 3. Applications Table
CREATE TABLE applications (
    application_id VARCHAR(64) PRIMARY KEY,             -- e.g., "APP-2026-INC-883921
    citizen_id UUID REFERENCES citizens(citizen_id),
    certificate_type VARCHAR(64) NOT NULL,              -- "INCOME_AND_ASSET", "DOMIC
    status VARCHAR(32) NOT NULL,                        -- "INTAKE", "AI_PROCESSING",
    client_document_sha256 VARCHAR(64) NOT NULL,
    minio_object_path VARCHAR(256) NOT NULL,
    assigned_officer_id VARCHAR(32) REFERENCES officers(officer_id),
    sla_escrow_deadline TIMESTAMP WITH TIME ZONE NOT NULL,
    sla_escalated BOOLEAN DEFAULT FALSE,
    fraud_risk_score NUMERIC(5,4),                      -- 0.0000 to 1.0000
    opa_policy_evaluation JSONB,
    created_at TIMESTAMP WITH TIME ZONE DEFAULT NOW(),
    updated_at TIMESTAMP WITH TIME ZONE DEFAULT NOW()
);

-- 4. Immutable Audit Log Table
CREATE TABLE immutable_audit_trail (
    audit_id BIGSERIAL PRIMARY KEY,
    application_id VARCHAR(64) NOT NULL,
    actor_id VARCHAR(64) NOT NULL,                      -- CitizenID, OfficerID, or "
    action_type VARCHAR(64) NOT NULL,                   -- "SUBMIT", "OCR_EXTRACT", "
    action_payload_hash VARCHAR(64) NOT NULL,           -- SHA-256(Action_Data)
    ip_address_hash VARCHAR(64),
    timestamp TIMESTAMP WITH TIME ZONE DEFAULT NOW()
);

-- Row-Level Security (RLS) Policy Example:
-- Officers can ONLY view applications assigned to their specific officer_id in their
ALTER TABLE applications ENABLE ROW LEVEL SECURITY;

CREATE POLICY officer_assigned_applications_only
    ON applications
    FOR SELECT
    USING (assigned_officer_id = current_setting('request.jwt.officer_id', true));
```

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2. Permissioned Blockchain Ledger Data Structures (Go Chaincode)

```
package main

import "time"

// CertificateState represents the on-chain immutable proof commitment
type CertificateState struct {
    CertificateID        string    `json:"certificate_id"`        // "GOV-2026-INC-88
    CertificateType      string    `json:"certificate_type"`      // "INCOME_AND_ASSE
    StateHash            string    `json:"state_hash"`            // SHA-256(CertID |
    DocumentSHA256       string    `json:"document_sha256"`       // Client-side raw
    CitizenIdentityHash  string    `json:"citizen_identity_hash"` // SHA-256(National
    IssuerOfficerID      string    `json:"issuer_officer_id"`     // "OFFICER-REV-409
    IssuerPKIFingerprint string    `json:"issuer_pki_fingerprint"`// SHA-256(Officer
    OfficerDigitalSig    string    `json:"officer_digital_sig"`   // ECDSA Base64 Sig
    Status               string    `json:"status"`                // "VALID" or "REVO
    IssuedTimestamp      time.Time `json:"issued_timestamp"`
    BlockNumber          uint64    `json:"block_number"`
    MerkleRoot           string    `json:"merkle_root"`
}

// CertificateRevocationEvent logs immutable revocation
type CertificateRevocationEvent struct {
    CertificateID   string    `json:"certificate_id"`
    RevokedByOfficer string   `json:"revoked_by_officer"`
    RevocationReason string   `json:"revocation_reason"`
    StatutoryCitation string  `json:"statutory_citation"`
    Timestamp       time.Time `json:"timestamp"`
    Signature       string    `json:"signature"`
}
```

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3. Python FastAPI Models (Pydantic v2 Contracts)

```
from pydantic import BaseModel, Field
from typing import Dict, Any, Optional
from datetime import datetime

class DocumentUploadRequest(BaseModel):
    citizen_national_id_token: str = Field( ... , description="OIDC Verified Citizen To
```

```
    certificate_type: str = Field( ... , example="INCOME_AND_ASSET")
    client_document_sha256: str = Field( ... , min_length=64, max_length=64, example="4
    raw_file_base64: str = Field( ... , description="Encrypted payload file binary")

class AIProcessingResult(BaseModel):
    application_id: str
    ocr_confidence: float = Field( ... , ge=0.0, le=1.0)
    extracted_entities: Dict[str, Any]
    biometric_similarity_score: float = Field( ... , ge=0.0, le=1.0)
    fraud_tamper_score: float = Field( ... , ge=0.0, le=1.0)
    syndicate_cluster_flag: bool
    opa_policy_passed: bool
    opa_evaluation_details: Dict[str, Any]

class OfficerApprovalPayload(BaseModel):
    application_id: str
    officer_id: str
    decision: str = Field( ... , example="APPROVE") # APPROVE, REJECT, QUERY
    statutory_citation_code: Optional[str] = None
    officer_ecdsa_signature: str = Field( ... , description="Hardware HSM/PKI ECDSA Sig
    timestamp: datetime = Field(default_factory=datetime.utcnow)
```